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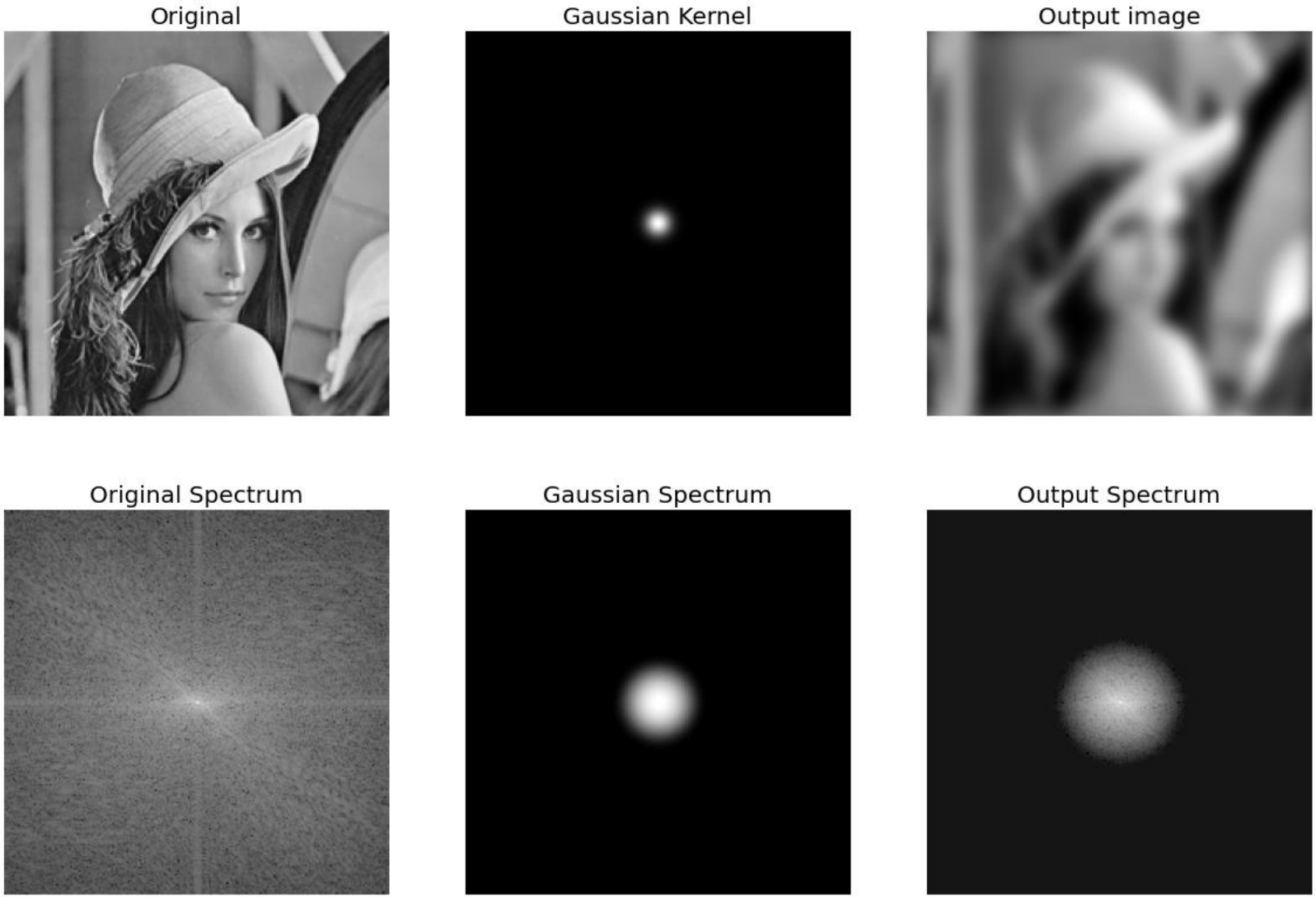
응용수치해석 및 실습 제 4주

권기운

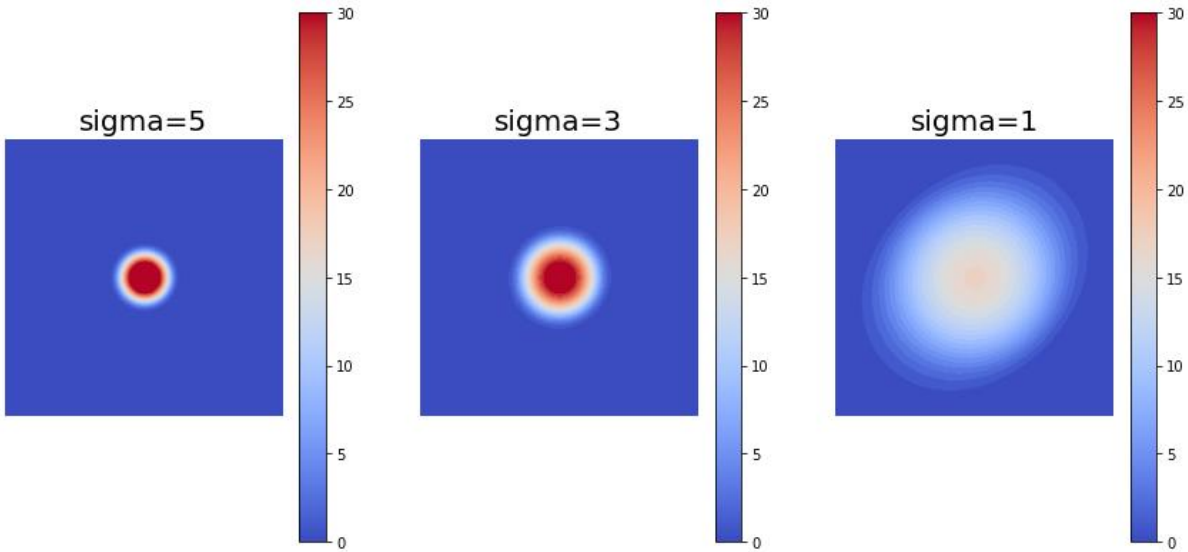
- Convolution 정리
- Frequency domain Gaussina blurr
- Frequency domain filtering
  - HPF
  - LPF
  - BPF
  - Notch filter

- 컨볼루션과 주파수 영역 필터링
- 컨볼루션 정리 및 주파수 영역 가우시안 흐림
- 주파수 영역 필터링(scipy ndimage 모듈 및 scikit-image 사용)
- 컨볼루션 정리 $f(x,y)*h(x,y)=F(u,v)H(u,v)$
- h 커널, 필터, 마스크, 윈도우, 열화 함수. 향상 함수

실습 1. 가우시안 필터 합성곱과 주파수 영역 변환

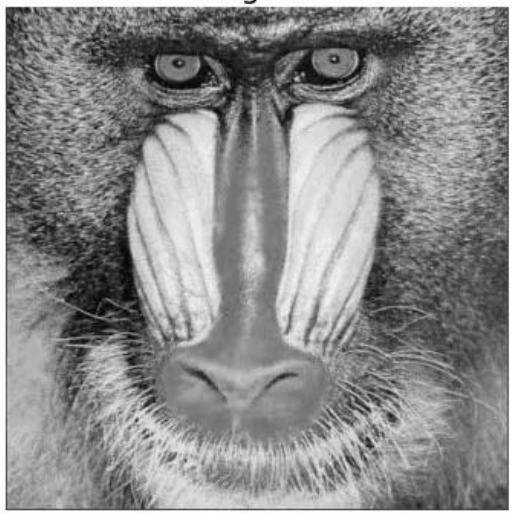


실습 2. Gaussian Low Pass Filter

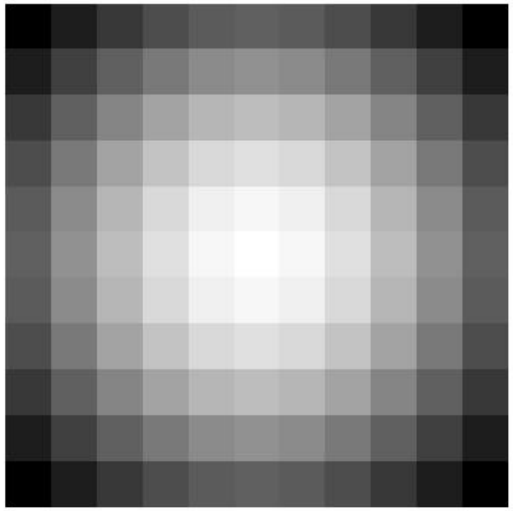


실습 3. scipy.signal.fftconvolve를 사용한 가우시안 필터

Original

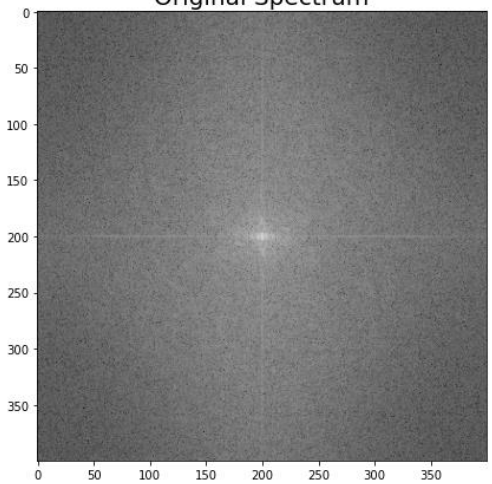


Gaussian Kernel

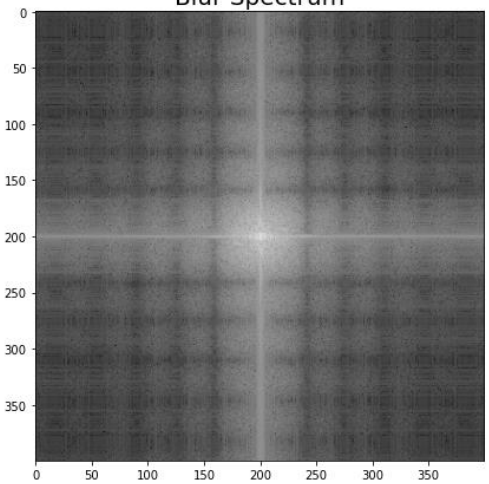


Blurred

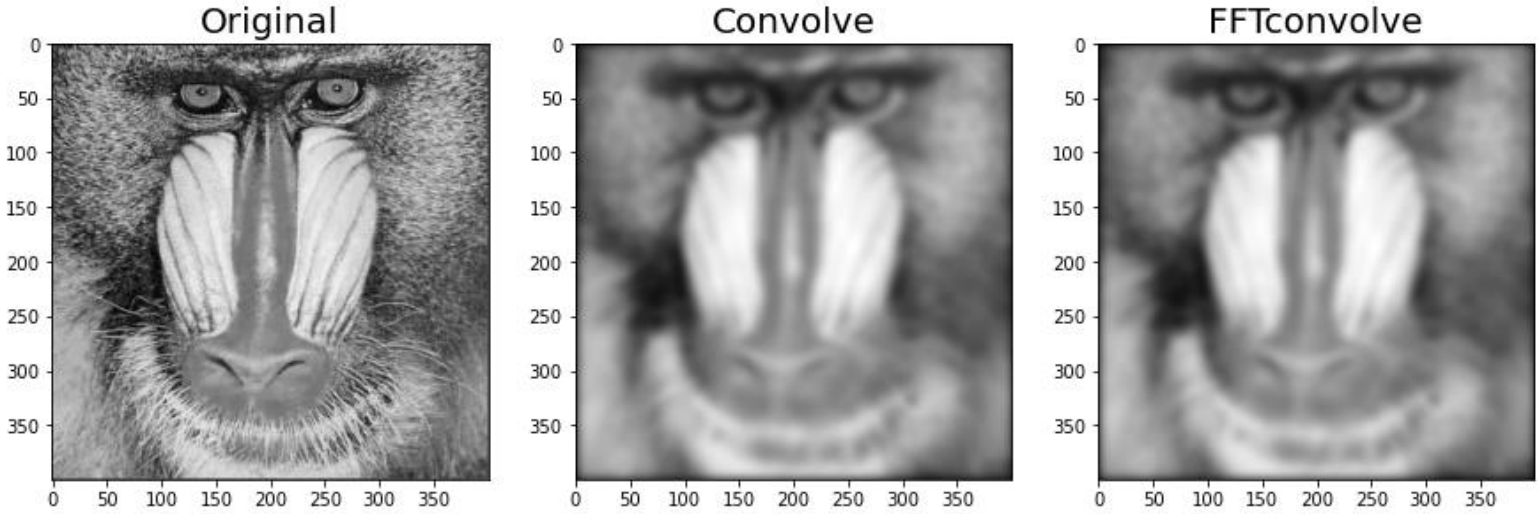
Original Spectrum

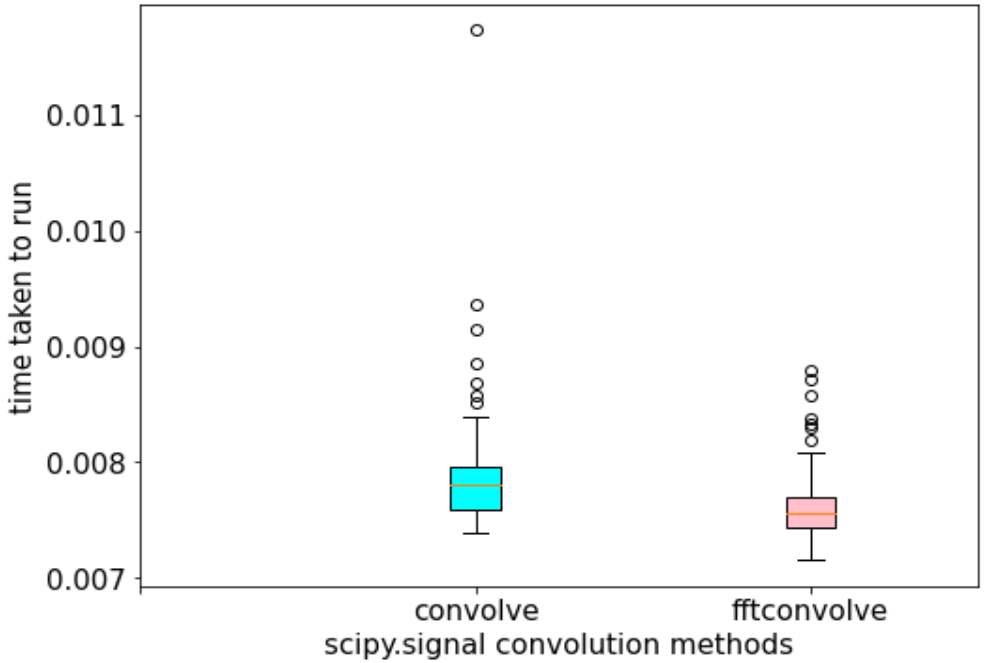


Blur Spectrum



실습 4. 시간 비교


Original
Convolve
FFTconvolve
0
50
100
150
200
250
300
350


time taken to run
0.011
0.010
0.009
0.008
0.007
convolve
fftconvolve
scipy.signal convolution methods

필터

- 필터의 효과
  - 대비(Contrast)를 향상(Enhancement)
  - 잡음 제거(Smoothing)
  - 알려진 패턴을 감지(Template Matching)

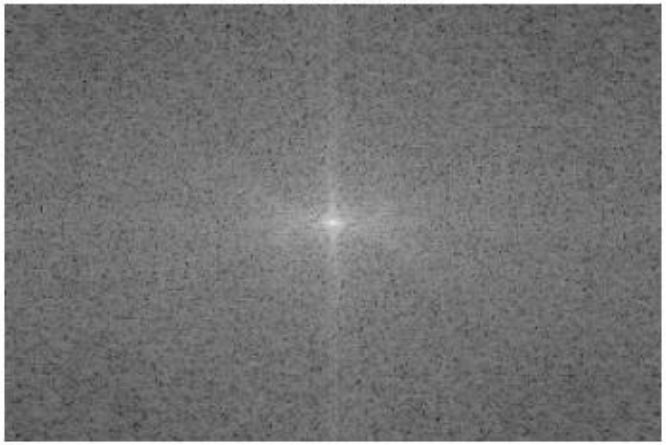
- 고역통과 필터 HPF, Highpass filter
  - 고주파 성분 edge, detail, noise

- HPF 구현
  - 1. scipy.fftpack.fft2로 2D FFT를 수행하고 영상의 주파수 영역표현 구함
  - 2. 고주파 성분 유지: 저주파 성분 제거
  - 3. 역 FFT를 수행하여 영상을 재구성

실습 5. High Pass Filtering

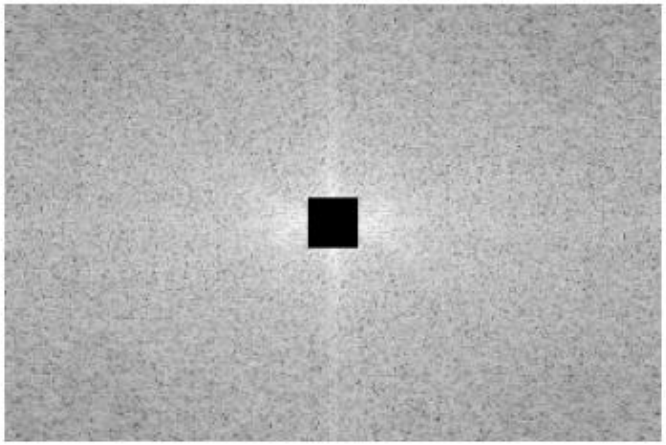
Original Image

Original Frequency

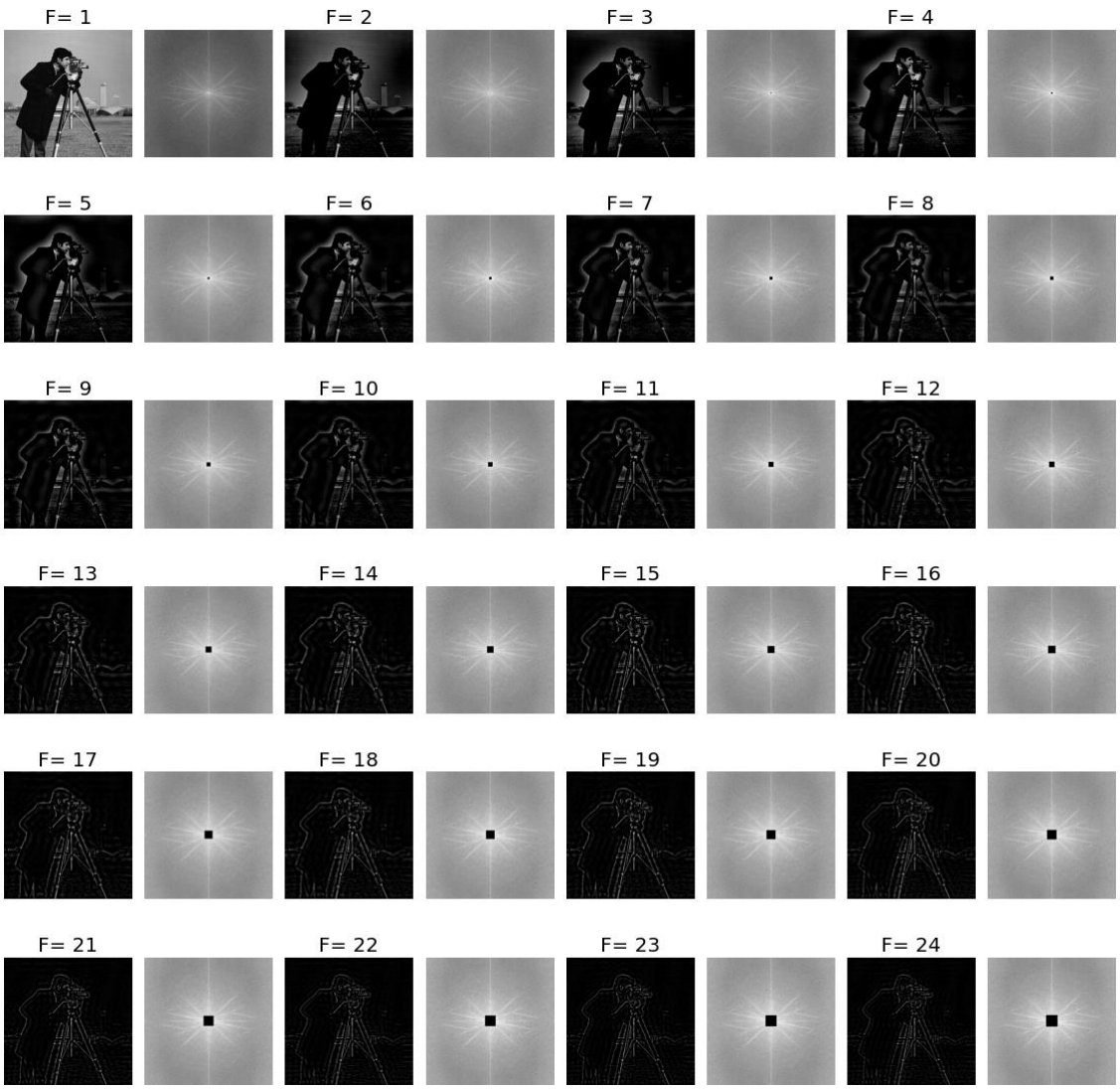


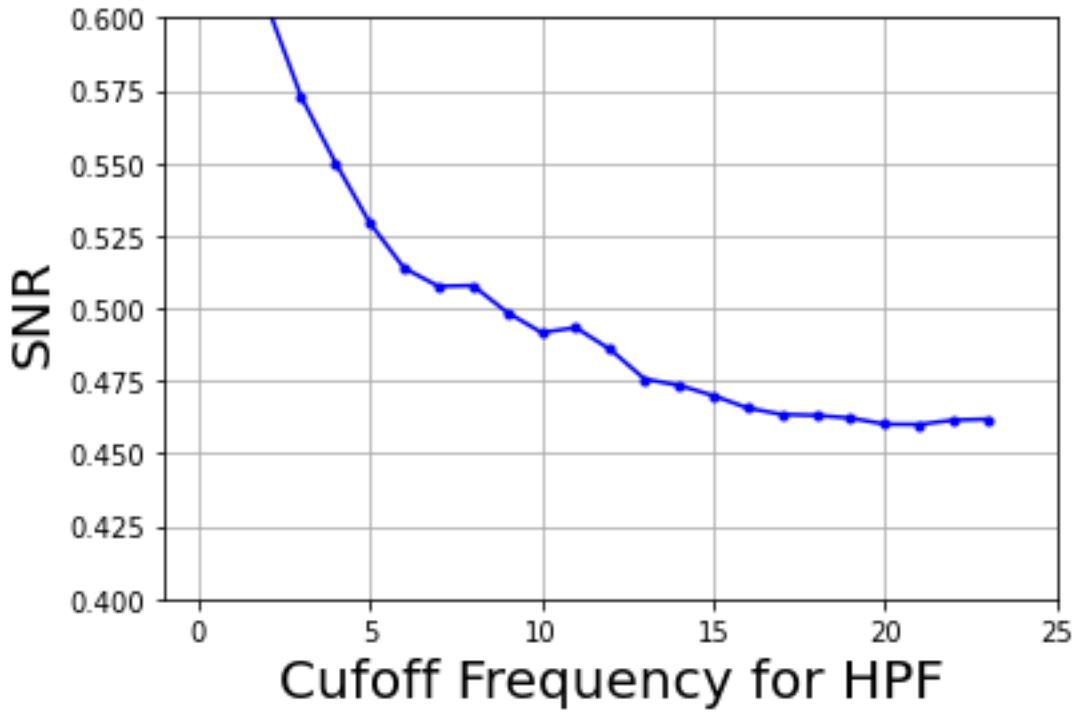
High Pass Image

High Pass Frequency



실습 6. HPF for 카메라맨





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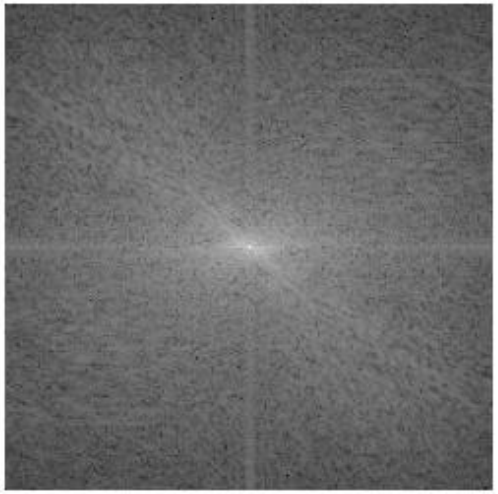
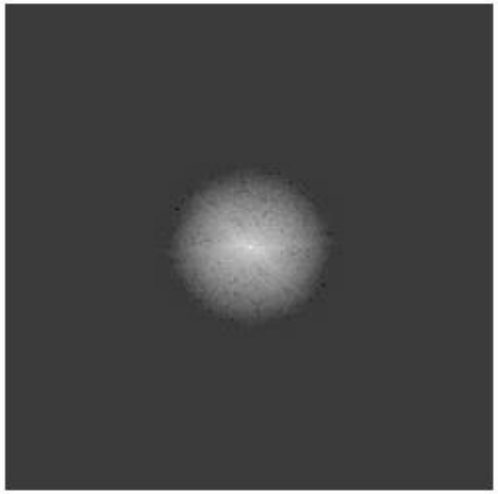
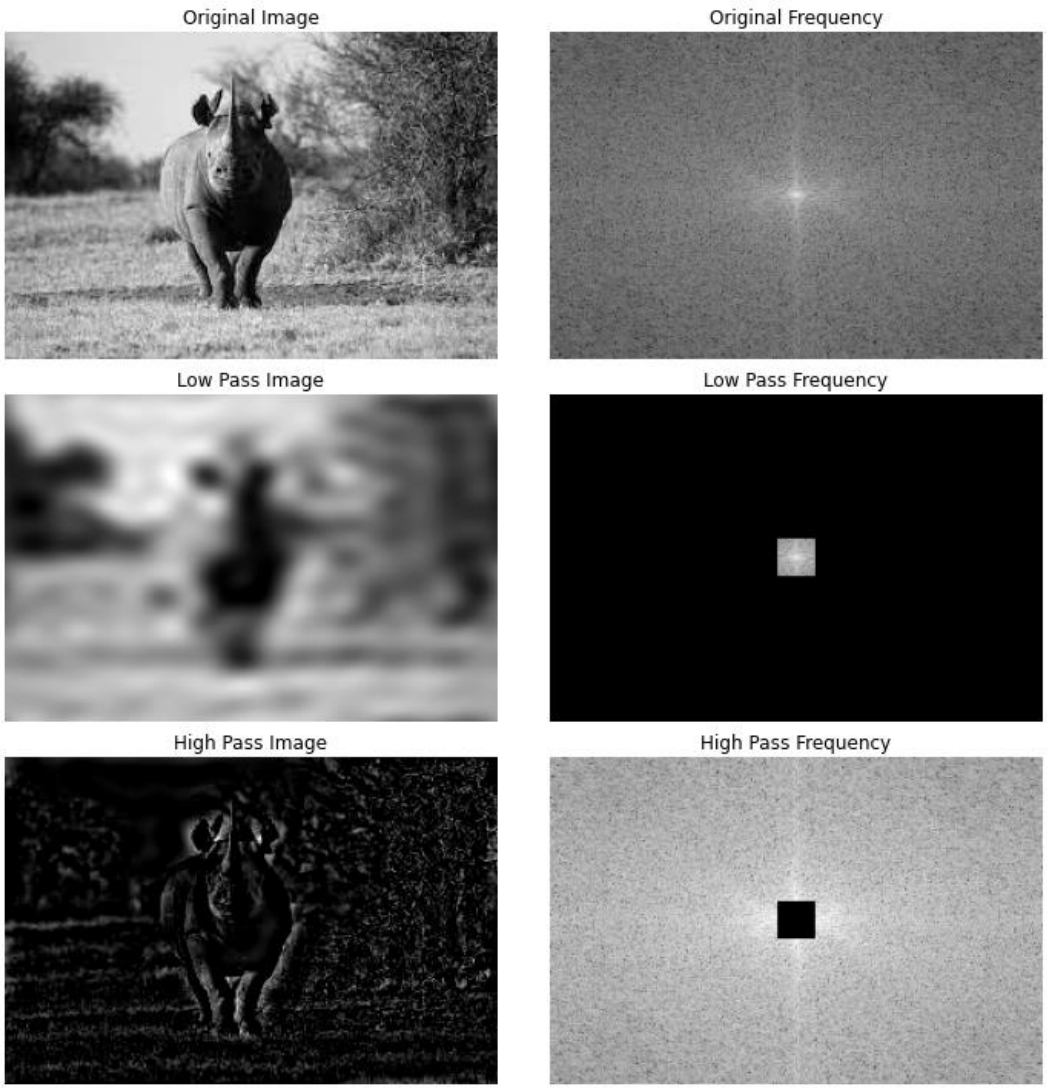
실습 7. Low Pass Filtering

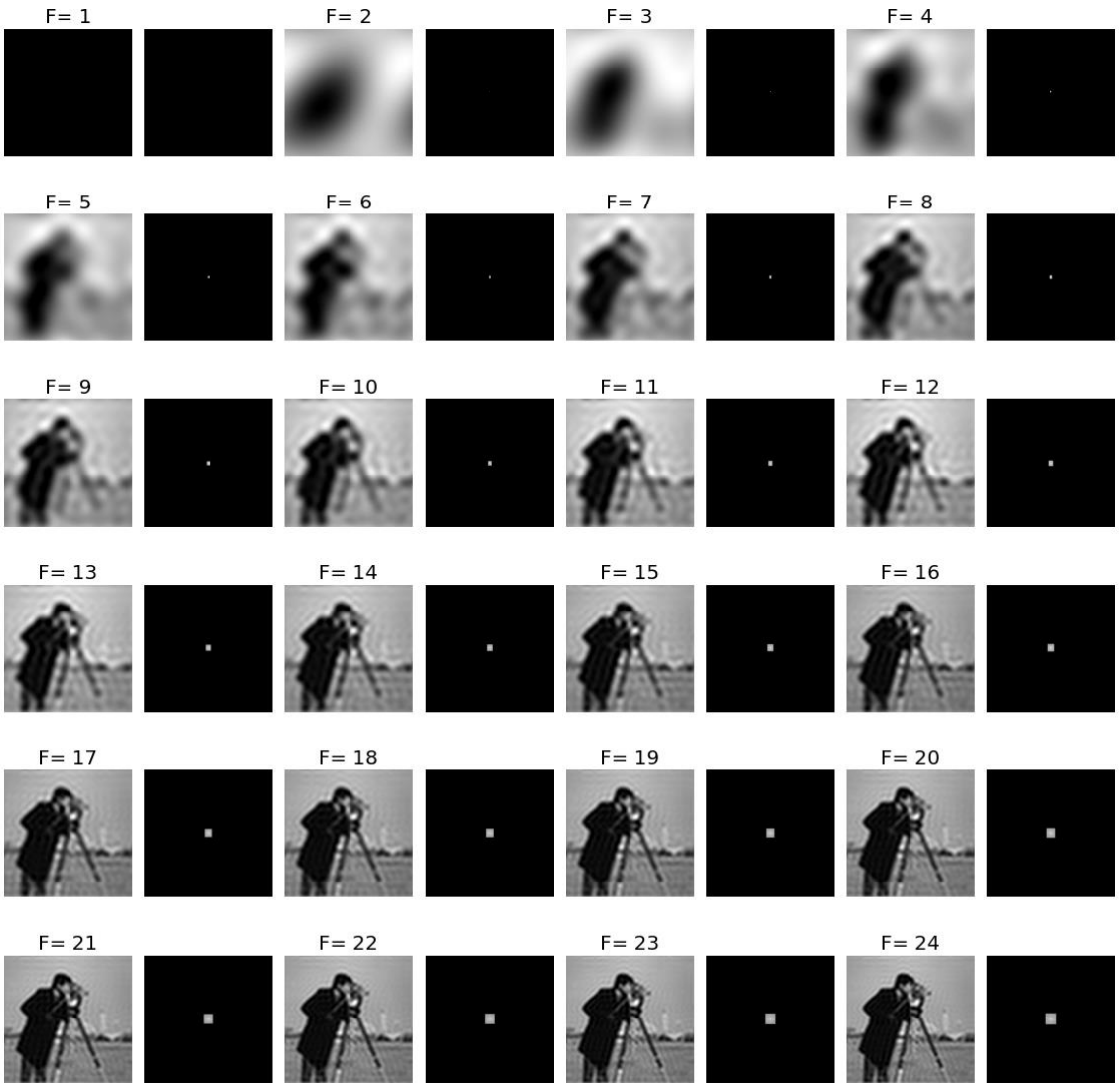




실습 8. 코뿔소 LPF, HPF



실습 9. 카메라맨 LPF


F= 1
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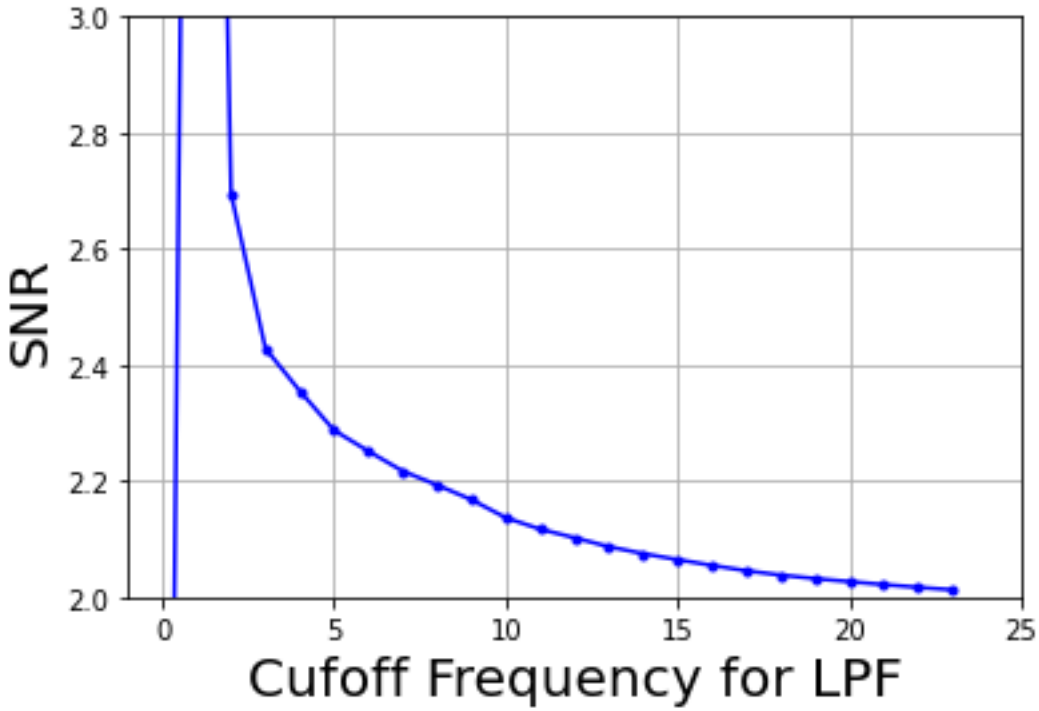
F= 4
F= 5
F= 6
F= 7
F= 8
F= 9
F= 10
F= 11
F= 12
F= 13
F= 14
F= 15
F= 16
F= 17
F= 18
F= 19
F= 20
F= 21
F= 22
F= 23
F= 24



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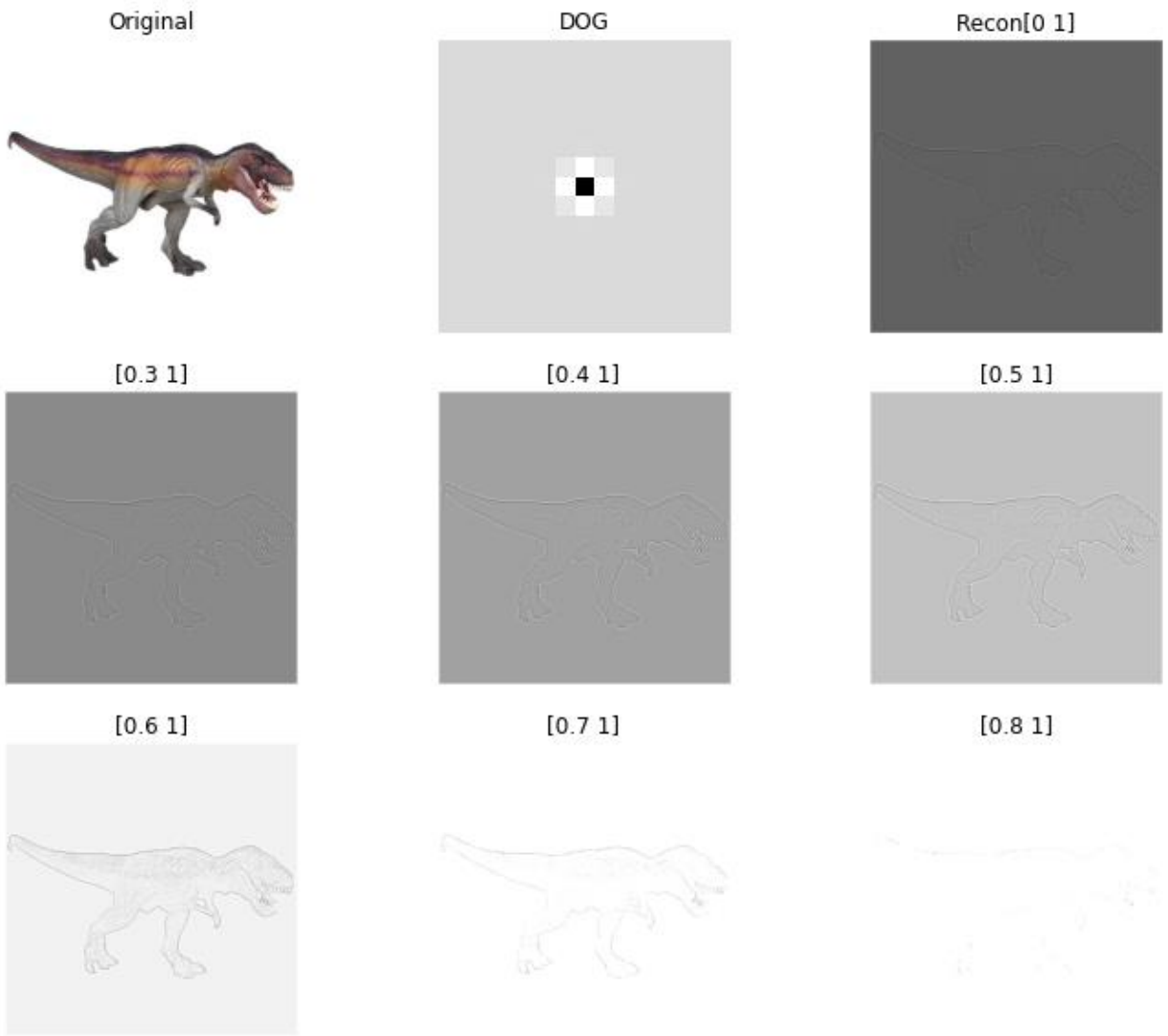
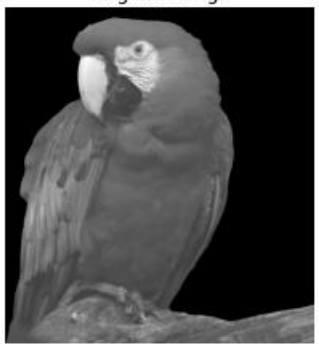
실습 10. 대역통과필터(BPF) : Difference of Gaussian(DOG)



실습 11. 대역차단 필터 Notch filter

Original Image



Original Image Spectrum

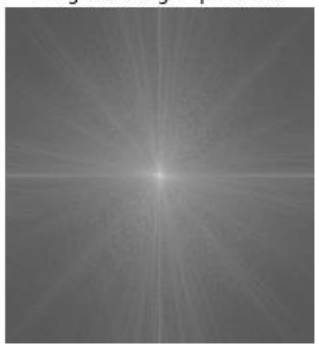
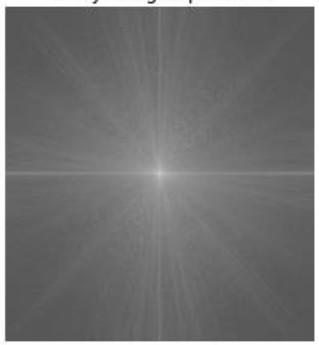
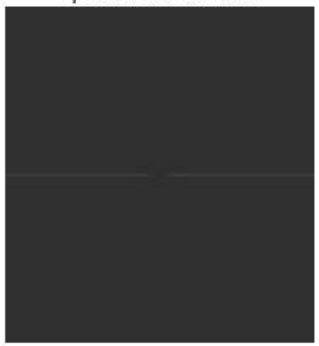


Image after adding Sinusoidal Noise

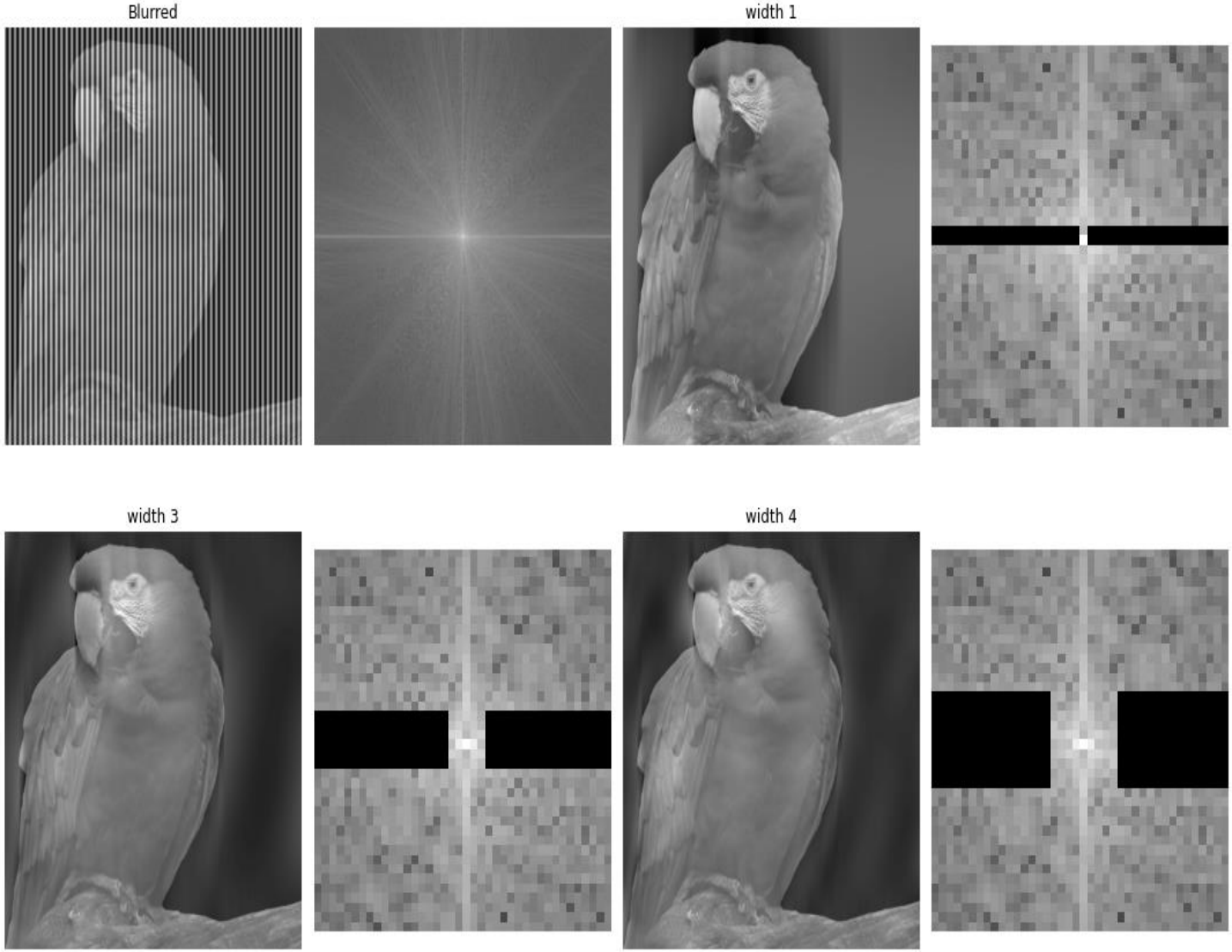
Noisy Image Spectrum



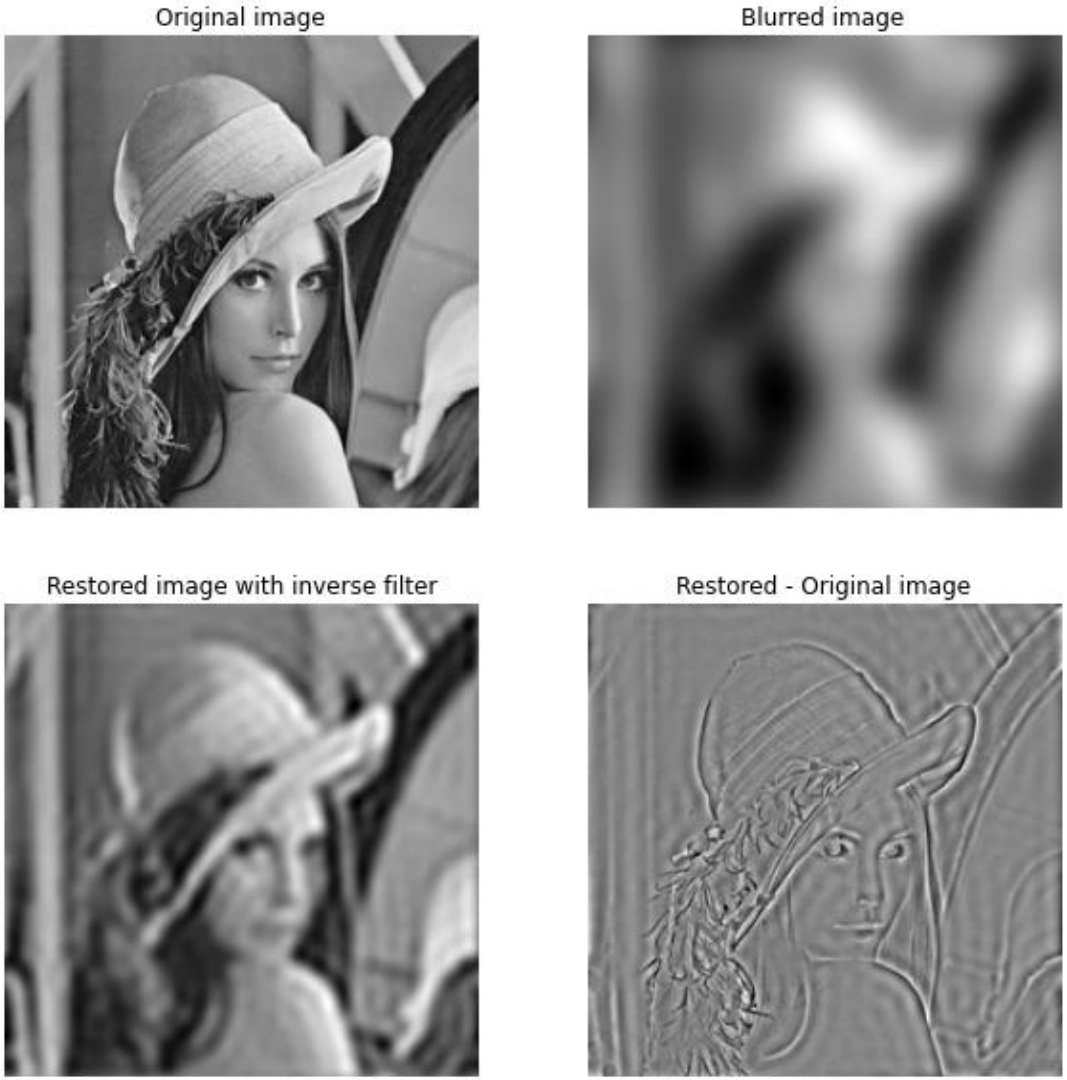
Spectrum Difference



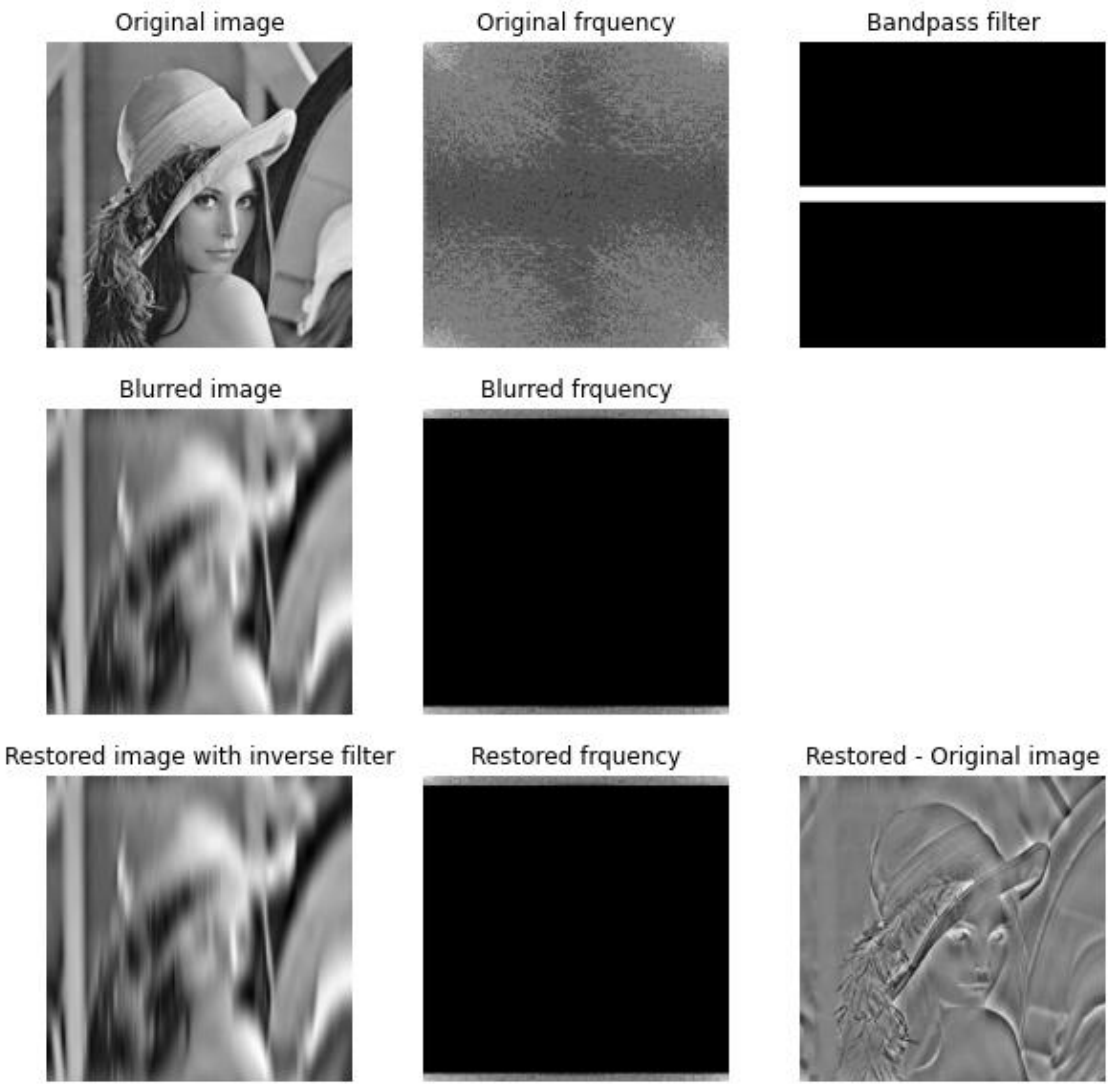
실습 12. 필터 설계



실습 13. 영상 복원, Deconvolution: Inverse filter



실습 14. Lena : Inverse filter



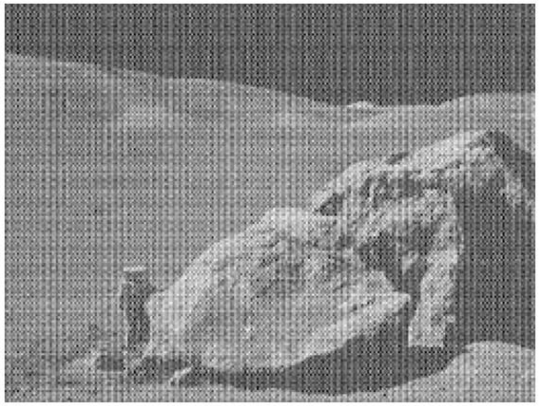
실습 15. 영상복원 Wiener filter

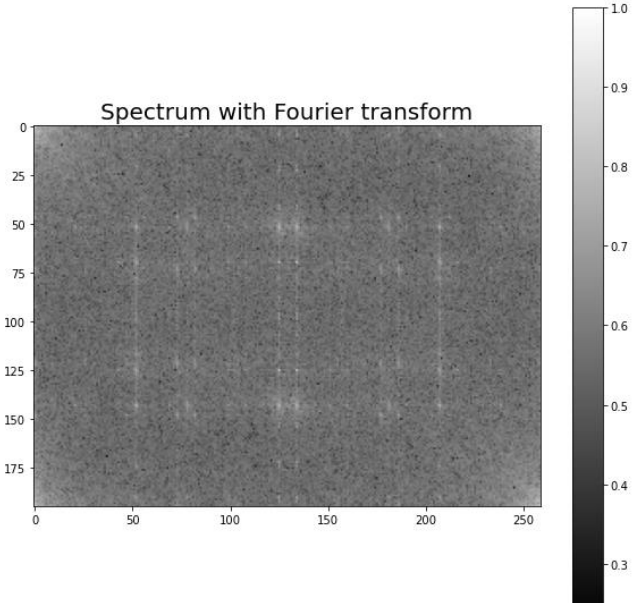
Original image

Noisy blurred image

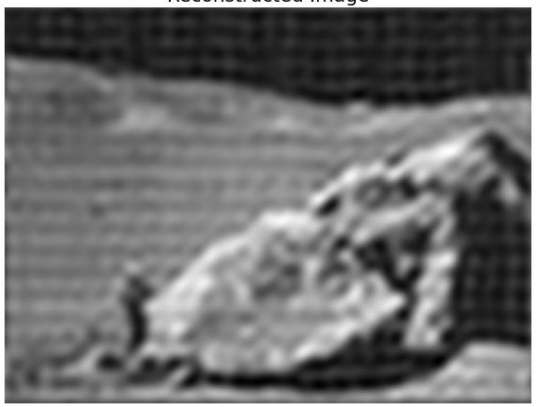
Self tuned restoration

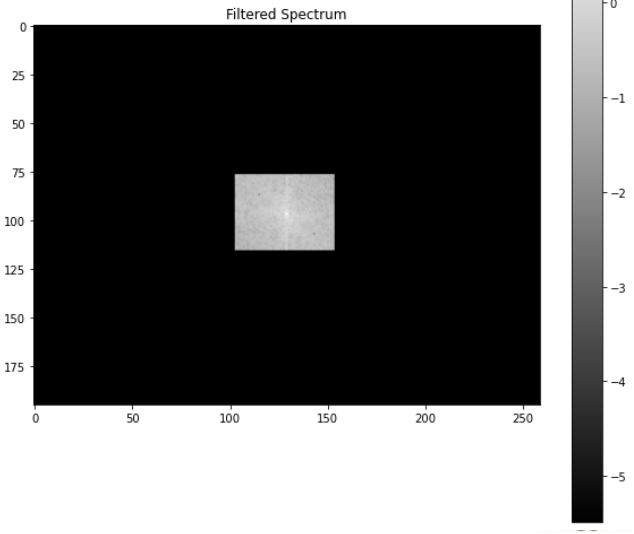
실습 16. 영상 잡음 제거





Reconstructed Image





결론

- 여러 가지 방법으로 합성곱과 주파수 변환과의 관계를 알아보았다.
- 고역필터(HPF), 저역필터(LPF), 대역통과필터(BPF), 대역저지필터(Notch filter)에 대해서 알아보았다.
- 주파수 변환을 이용하여 영상 잡음 제거, 영상복원(Deconvolution), 에지 검출을 수행할 수 있다.